HamidReza Saadi Dadmarzi

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Education

Ph.D. in Computational Sciences and Engineering, Koç University — Fall 2026–present
Advised by Assoc. Prof. Mehmet Cengiz Onbaşlı.

M.Sc. in Computer Science and Engineering, Koç University — 2023–2026
Thesis Secure Single Password Authentication Protocols in Distributed Systems.
Advised by Prof. Alptekin Küpçü. Defended 28 September 2026.

M.Sc. in Pure Mathematics, Amirkabir University of Technology — 2017–2020

Minor in Physics, Amirkabir University of Technology — 2016–2018

B.Sc. in Applied Mathematics, Amirkabir University of Technology — 2014–2017

Research

- **Post-quantum cryptography and quantum information.** AEGIS–VAKE for protocol assurance and Lean proof reuse, timing leakage in post-quantum implementations, and approximately mutually unbiased bases in $\mathbb{C}^6$.
- **Privacy-preserving ML and ML theory.** PersonGuard and CrossLead extend *Structure Matters* to person-level ECG privacy. Grassmann/PCA research studies machine unlearning.
- **Secure multiparty computation and theoretical cryptography.** Designed and benchmarked threshold password authentication in UpSPA, browser integration in SPA-Web, and quorum recovery in RoUpSPA.

Selected Publications and Manuscripts

- **Structure Matters: Differentially Private Spectral Statistics for ECG Time Series**.
  Accepted poster, TS-LIMITS workshop at NeurIPS 2026.
- **SPA-Web: Browser-Based Single-Password Authentication**.
  Accepted conference paper, IEEE ISC Türkiye 2026.
- **UpSPA: Secure and Updatable Single Password Authentication**.
  IACR ePrint 2026/784. Submitted to ACNS 2027 (Cycle 1).
- **RoUpSPA: Robust Updatable Single Password Authentication**.
  Preprint prepared for ACM Transactions on Privacy and Security (TOPS).

Research and Professional Experience

Koç University — 2023–2026 (M.Sc. research)
Research Assistant, KU-Crypto Lab. Designed and evaluated distributed authentication protocols.
CURE implementation support in Go/Lattigo for privacy-preserving split learning.

BTL Coin Software Developer. Crypto-asset tools in Go and Docker. — 2021–2023

Institute for Research in Fundamental Sciences (IPM) Quantum ML research. — 2020–2021

Teaching, Mentoring, and Service

Koç University teaching assistant. Modern Cryptography (Fall 2026 and Fall 2025), Computer Networks & Security (Spring 2026 and Spring 2025).

Mentored five interns. Outstanding Teaching Assistant Award, Koç University (2025). Reviewer for IEEE TIFS, ACM ASIACCS, ESORICS, and ACNS.

Technical Skills

Rust, Go, Python, MATLAB, Lean 4, Docker, Qiskit, PyQuil, and Lattigo.